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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion. (10 Marks)

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard

Referencing Style for all citations and references.

For essay -style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided. Include a signed
declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: M.K MORE

Date: 31 October 2025

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Question 2:

2.1

CEO Fraud, also known as Business Email Compromise (BEC), is a targeted scam where criminals pose as senior executives to persuade employees into transferring money or disclosing sensitive data.

Common Tactics Include:

- Spoofed or look-alike emails: Attackers forge addresses or use domains that closely resemble the companies.
- Manipulative messaging: They often create urgency or confidentiality to pressure quick compliance
- Compromised real accounts: In advanced cases, scammers gain access to legitimate executive inboxes, making the deception more convincing.

Prevention Strategies:

- Implement email authentication (SPF, DKIM, DMARC).
- Always verify financial or data requests via a secondary channel.
- Protect accounts with multi-factor authentication.
- Provide continuous staff training and simulated phishing exercises.
- Use dual approval for large transactions or vendor changes.
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Therefore, preventing CEO fraud depends on combining strong technology safeguards, internal verification processes and ongoing employee awareness.

2.2

Sam Dunn Enterprises has rules to keep company information and technology safe, private, and available for the people who need it. This policy applies to everyone who uses company systems, including employees, contractors, and freelancers, whether they work in the office or from home.

Company data falls into three types: Restricted (very private, like financial or personal records), Confidential (for internal use only), and Public (safe to share outside the company). Work on computers and accounts should be used mainly for business, though a little personal use is fine if it doesn't cause problems.

Only software that IT has approved can be installed, and people should have access only to the tools they need for their jobs. Passwords must be long and strong—at least 14 characters—and changed regularly. Important systems require multi-factor authentication for extra protection. Sensitive information should only be sent through secure, encrypted channels, and any strange emails or requests should be reported to IT right away.

All company devices must have antivirus protection, encryption, and use the VPN when connecting remotely. If something seems wrong, like a suspected hack or virus, report it immediately and don't try to fix it yourself. Breaking these security rules can lead to disciplinary action, and the company reviews the policy every year to keep it up to date.

2.3

When a user fills out a form on a website, PHP collects the information using special built-in arrays called superglobals. The two most common are `$_GET` and `$_POST`, which store data depending on how the form was sent. If the form uses the POST method, the data is hidden in the request body and doesn't appear in the URL—this makes it better for private or sensitive information. If the form uses the GET method, the data is attached to the URL as a query string, which is fine for things like searches or filters that aren't private.

In your PHP script, you can check which method was used and pull the values safely, for example with `$_POST['user_name']`. It's very important to protect your code from common attacks. To stop XSS, use `htmlspecialchars()` before showing user input. To prevent SQL injection, always use prepared statements instead of putting user data directly into queries. Avoid CSRF attacks by adding a hidden token in forms and checking it when data is submitted. Also, always validate and sanitize input on the server to make sure users don't send harmful or incorrect data.

When many people submit forms at the same time, your database can get overloaded. To handle this smoothly, developers often use background systems like message queues (for example, RabbitMQ or Redis) so forms are processed one step at a time. By combining good coding practices, careful validation, and proper security, your PHP forms can stay safe, reliable, and ready to handle real-world traffic.

Reference:

- Proofpoint. (2021). *What Is CEO Fraud?* [online] Available at: <https://www.proofpoint.com/us/threat-reference/ceo-fraud> [Accessed 20 Oct. 2025].
- Hornetsecurity (2023). *The Top 5 Spear Phishing Examples and Their Psychological Triggers - Hornetsecurity – Next-Gen Microsoft 365 Security*. [online] Hornetsecurity – Next-Gen Microsoft 365 Security. Available at: <https://www.hornetsecurity.com/en/blog/spear-phishing-examples/#:~:text=By%20impersonating%20a%20person%20of,to%20take%20action%20without%20question>. [Accessed 20 Oct. 2025].
- Gabil, D. (2024). PHP: Superglobals \$_GET, \$_POST, \$_REQUEST, \$_COOKIE, \$_SESSION. [online] ScriptVerse. Available at: <https://scriptverse.academy/tutorials/php-superglobals.html> [Accessed 25 Oct. 2025].
- Qmul.ac.uk. (2025). Secure Email Transmission of Personal Data Using Encryption - IT Services. [online] Available at: <https://www.qmul.ac.uk/its/cybersecurity/email-security/email-encryption/> [Accessed 25 Oct. 2025].
- chrisda (2025). Use DMARC to validate email, setup steps - Microsoft Defender for Office 365. [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/defender-office-365/email-authentication-dmarc-configure> [Accessed 26 Oct. 2025].